

PerCDL: Personalized Convolutional Dictionary Learning of Physiological Time Series

Aymane Hamdaoui
aymane.hamdaoui@telecom-paris.fr
ENS Paris-Saclay
Paris, France

Titouan Duhazé
titouan.duhaze@telecom-paris.fr
ENS Paris-Saclay
Paris, France

Abstract

Physiological time series, such as locomotion or ECG signals, exhibit a dual structure: global patterns shared across a population and local variations specific to individuals. Standard Convolutional Dictionary Learning (CDL) approaches often struggle to reconcile these aspects, either ignoring individual variability (Population CDL) or overfitting to individuals (Individual CDL). In this project, we study and implement Personalized CDL (PerCDL), a framework introduced by Roques et al. that models local information as a personalized spatiotemporal transformation of a global dictionary. We reproduce the authors' method, specifically focusing on Time Warping transformations, and evaluate its performance on both synthetic data and real-world human locomotion datasets. Our experiments confirm that PerCDL successfully separates shared structures from individual idiosyncrasies, offering superior interpretability and reconstruction compared to baseline methods.

1 Introduction

Dictionary Learning (DL) and its shift-invariant counterpart, Convolutional Dictionary Learning (CDL), are powerful unsupervised methods for decomposing time-series data into interpretable motifs (atoms) and sparse activations. These methods are particularly well-suited for physiological signals, which often consist of repeated events (e.g., heartbeats, gait cycles).

However, a major challenge in analyzing such data is the "local-global" structure. Anatomical similarities create shared patterns across a population (global), while pathology or biomechanical differences induce specific variations (local). Standard CDL applied to the whole population (PopCDL) tends to treat individual variations as noise. Conversely, applying CDL to each subject independently (IndCDL) fails to capture shared structures and is susceptible to noise.

To address this, we explore the **Personalized CDL (PerCDL)** framework proposed by Roques et al. (2025). PerCDL learns a unique set of global atoms and a set of learnable transformation functions (parameterized by "personalization parameters") that map the global atoms to individual-specific forms. The method focuses on Time Warping (TW) as the primary transformation to handle temporal misalignment and shape variability.

Contributions and Work Distribution. Unlike the original paper which relies on JAX, we chose to perform a **complete re-implementation of the PerCDL framework using PyTorch**. This approach allowed us to fully grasp the optimization mechanics, although it introduced specific computational challenges compared to the original implementation.

Furthermore, instead of the public dataset used in the article, we evaluated our implementation on a **clinical gait dataset** provided by medical partners from the "Immersion à l'Hôpital" course.

- **Implementation:** We ported the meta-algorithm (CSC, IPU, PerCDU) and the differentiable Time Warping operators to PyTorch from scratch.
- **Data:** We processed a cohort of 19 patients. The data is multivariate with 9 channels corresponding to 3 sensors placed on the Right Foot, Left Foot, and Trunk.
- **Work Distribution:** We collaborated on the mathematical derivations and the overall project design. For the experimental phase, Titouan focused on the implementation of the core algorithm and the plot functions. Meanwhile, Aymane worked on developing tools for comparisons and preprocessing. Both also worked on the analysis of the code provided by the authors to deepen their understanding.

2 Method

2.1 Standard CDL (Initialization)

Let $X \in \mathbb{R}^{S \times N \times P}$ be a dataset of S time series, where each signal x^s has length N and P channels. The standard Convolutional Dictionary Learning (CDL) problem assumes that all signals share a common set of K multivariate atoms $\Phi \in \mathbb{R}^{K \times L \times P}$.

Crucially, each atom ϕ_k is a collection of P temporal patterns (one per channel) that are locked in time. A single activation code z_k^s triggers the presence of atom k across all channels simultaneously. This enforces temporal coupling between channels: if an event occurs, it appears on all sensors with a fixed relative shape.

Mathematically, for a specific channel $p \in \{1, \dots, P\}$, the signal is reconstructed as:

$$x^s[\cdot, p] \approx \sum_{k=1}^K z_k^s * \phi_k[\cdot, p] \quad (1)$$

In the initialization phase, we obtain the global dictionary via alternating minimization:

- **Convolutional Sparse Coding (CSC):** We fix Φ and solve for the sparse activations Z . Note that the reconstruction error is summed over all channels, ensuring the activation z_k^s explains the data across all dimensions jointly:

$$Z \leftarrow \arg \min_{Z \geq 0} \sum_{s=1}^S \sum_{p=1}^P \left\| x^s[\cdot, p] - \sum_{k=1}^K z_k^s * \phi_k[\cdot, p] \right\|_2^2 + \lambda \|Z\|_0 \quad (2)$$

- **Convolutional Dictionary Update (CDU):** We fix Z and update the multivariate atoms Φ :

$$\Phi \leftarrow \arg \min_{\Phi: \|\phi_k\|_2=1} \sum_{s=1}^S \sum_{p=1}^P \left\| x^s[\cdot, p] - \sum_{k=1}^K z_k^s * \phi_k[\cdot, p] \right\|_2^2 \quad (3)$$

2.2 Personalized CDL (PerCDL)

The core contribution of this work is the Personalization phase. PerCDL assumes that the observed signals are generated by *personalized atoms*, denoted as $\hat{\Phi}$, which are derived from the global atoms Φ via a transformation function f .

For a transformation $f : (\mathbb{R}^P)^L \times \mathbb{R}^M \rightarrow (\mathbb{R}^P)^L$, the personalized atom for subject s and pattern k is defined as $\hat{\phi}_k^s = f(\phi_k, a_k^s)$, where $a_k^s \in \mathbb{R}^M$ are the personalization parameters.

The PerCDL optimization problem is formulated as:

$$\min_{Z, \Phi, A} \sum_{s=1}^S \left\| x^s - \sum_{k=1}^K z_k^s * f(\phi_k, a_k^s) \right\|_2^2 + \lambda \|Z\|_0 \quad (4)$$

To solve this non-convex problem, we employ a three-step block coordinate descent scheme:

- (1) **Individual Parameters Update (IPU):** We fix the common dictionary Φ and activations Z to learn the personalization parameters A . This step fits the warp/transformation to the specific subject:

$$A \leftarrow \arg \min_A \sum_{s=1}^S \left\| x^s - \sum_{k=1}^K z_k^s * f(\phi_k, a_k^s) \right\|_2^2 \quad (5)$$

- (2) **CSC on Personalized Atoms:** We define the personalized dictionary $\hat{\Phi} = \{f(\phi_k, a_k^s)\}_{k,s}$. We then update the activations Z by solving the standard CSC problem using $\hat{\Phi}$ instead of Φ .
- (3) **Personalized Dictionary Update (PerCDU):** Finally, we update the shared global dictionary Φ . This step ensures that Φ remains the “centroid” or base shape from which all individual atoms are derived:

$$\Phi \leftarrow \arg \min_{\Phi: \|\phi_k\|_2=1} \sum_{s=1}^S \left\| x^s - \sum_{k=1}^K z_k^s * f(\phi_k, a_k^s) \right\|_2^2 \quad (6)$$

2.3 Time Warping Transformation

The specific transformation used is a **Time Warp**. The warped atom is defined as $\tilde{\phi}(t) = \phi(\psi_a(t))$, where ψ_a is a time-warping function. Following the paper, we use a diffeomorphism parameterization based on a composition of D displacements, composed of sine bases of width W . This ensures the transformation is smooth and invertible (non-degenerate). The hyper-parameters D (depth) and W (width) control the complexity and granularity of the allowed deformations.

3 Implementation Details

To solve the optimization problem efficiently while strictly enforcing the non-overlapping constraint, we deviated from standard gradient-based sparse coding. Instead, we implemented specific solvers for the CSC and Dictionary Update steps.

3.1 Exact ℓ_0 -CSC via Dynamic Programming

For the sparse coding step, we adapted the method proposed by [Truong and Moreau 2024]. The goal is to find the activations Z that minimize the reconstruction error under an ℓ_0 sparsity penalty, **with the constraint that atoms cannot overlap**:

$$\min_Z \left\{ \frac{1}{2} \|X - \Phi * Z\|_2^2 + \lambda \|Z\|_0 \right\} \quad (7)$$

We implemented this using a Dynamic Programming (DP) approach:

3.1.1 Pre-computation. We reshape the tensor dictionary Φ into a matrix $D \in \mathbb{R}^{(LP) \times K}$ where each column is a flattened kernel $\text{vec}(\phi_k)$. We then pre-compute the projection operator $\tilde{P}$, which is the regularized pseudo-inverse of D :

$$\tilde{P} = (D^\top D + \epsilon I)^{-1} D^\top \in \mathbb{R}^{K \times (LP)} \quad (8)$$

3.1.2 Local Cost Calculation. For every possible time window t in signal X^s , we define the local window $y_t = \text{vec}(X^s[t : t + L, \cdot])$. We compute the optimal local coefficients $u_t = \tilde{P} y_t$ assuming a single activation occurs at t . The local cost c_t of placing an atom at t is defined as the trade-off between the sparsity penalty and the reconstruction gain:

$$c_t = \lambda - \|D u_t\|_2^2 \quad (9)$$

3.1.3 Dynamic Programming Recursion. To strictly enforce the no-overlap constraint, we compute the accumulated cost vector $V \in \mathbb{R}^N$. $V[t]$ represents the minimum cost to explain the signal up to time t :

$$V[t] = \min(V[t-1], V[t-L] + c_{t-L}) \quad (10)$$

Here, the term $V[t-L] + c_{t-L}$ implies placing a new atom at $t-L$ (which ends at t), preventing any other atom from existing in the interval $(t-L, t]$.

3.1.4 Backtracking. Finally, we trace back the optimal path from $V[N]$. Whenever $V[t] = V[t-L] + c_{t-L}$, we record an activation at $t-L$ with coefficients u_{t-L} and jump back L steps. This guarantees an exact solution to the non-overlapping ℓ_0 problem [Truong and Moreau 2024].

3.2 PerCDU via Numerical Inverse Warping

Updating the common dictionary Φ is challenging because the transformation $f(\phi_k, a_k^s)$ introduces complex non-linearities (time warping) that make standard gradient descent unstable. To address this, we implemented a closed-form update based on **inverse warping**.

3.2.1 Patch Alignment. The core idea is to map the observed signal patches back to the “canonical” timeframe of the common atom. For every non-zero activation $z_k^s(t)$, we extract the corresponding signal patch $y = X^s[t : t + L, \cdot]$.

The forward warp is defined as $\psi(t) = t + \sum a_m b_m(t)$. Since this function is strictly monotonic (diffeomorphism), it is invertible but has no closed-form inverse. We approximate the inverse ψ^{-1} numerically:

- (1) Compute the forward warp $\psi(t_{grid})$ on a dense grid.

- (2) Interpolate to find the canonical time points $t_{\text{canonical}} = \psi^{-1}(t_{\text{warped}})$.
- (3) Resample the patch y at $t_{\text{canonical}}$ to obtain the aligned patch $\text{aligned}_{k,t}^s$.

3.2.2 Closed-Form Update. Once all patches are aligned to the canonical frame, the optimal common atom $\tilde{\phi}_k$ is simply the weighted average of these aligned patches. We include a regularization term λ_{reg} to prevent instability when an atom has few activations:

$$\tilde{\phi}_k = \frac{\sum_{s,t} z_k^s(t) \cdot \text{aligned}_{k,t}^s + \lambda_{\text{reg}} \phi_k^{\text{old}}}{\sum_{s,t} z_k^s(t) + \lambda_{\text{reg}}} \quad (11)$$

3.2.3 Stabilization. To ensure smooth convergence of the alternating minimization, we blend the new estimate with the previous dictionary using a momentum factor α :

$$\phi_k^{\text{new}} \leftarrow \text{normalize} \left((1 - \alpha) \phi_k^{\text{old}} + \alpha \tilde{\phi}_k \right) \quad (12)$$

This approach avoids the local minima often encountered when optimizing Φ directly through the warping function using automatic differentiation.

4 Data and Preprocessing

To evaluate the PerCDL framework in a realistic clinical context, we utilize a proprietary dataset of human locomotion provided by medical partners from the “Immersion à l’Hôpital” course. This section details the dataset characteristics, our diagnostic analysis of the signal quality based on spectral properties, and the resulting preprocessing pipeline.

4.1 Dataset Description

The dataset consists of multivariate physiological time series recorded from a cohort of $S = 19$ patients performing gait exercises. The data was acquired using Inertial Measurement Units (IMUs) placed at three anatomical locations: the Right Foot, the Left Foot, and the Trunk. Each sensor records 3-axis acceleration and/or angular velocity, resulting in $P = 9$ synchronous channels per subject with a sampling frequency $f_s = 95.95$ Hz.

The signals exhibit the “local-global” structure central to our study: while all subjects share the fundamental biomechanics of walking (global quasi-periodicity), significant variations exist due to individual gait pathologies and pacing (local variability).

4.2 Data Diagnosis

Prior to dictionary learning, we diagnosed the signal quality to distinguish between the physiological information and the sensor noise. We relied on the Additive White Gaussian Noise (AWGN) model introduced in the lectures, where the observed signal $y[n]$ is modeled as the clean physiological signal $x[n]$ corrupted by an additive noise component $b[n]$.

Spectral Analysis. To separate the signal from the noise, we analyzed the data in the frequency domain. As established in the course, adding white noise to a signal results in a constant addition to its power spectrum (a “noise floor”). Visual inspection of the Power Spectral Density (PSD) confirms this structure: high-energy peaks below 10 Hz (signal) and a flat, low-energy plateau above 20 Hz (noise).

Adaptive Bandwidth Estimation. In the lecture examples, the noise variance σ^2 was known, allowing for a direct cutoff calculation based on the noise floor level. However, for this clinical dataset, the exact sensor noise variance is unknown and variable across subjects.

Therefore, rather than assuming a fixed noise floor, we employed a **Cumulative Power** approach. This method assumes that the high-amplitude physiological signal accounts for the vast majority of the total energy. We define the cutoff frequency $f_c^{(s,p)}$ as the point where 95% of the spectral energy is captured:

$$f_c^{(s,p)} = \min \left\{ f \mid \frac{\int_0^f \text{PSD}_{s,p}(v) dv}{\int_0^{f_s/2} \text{PSD}_{s,p}(v) dv} \geq 0.95 \right\} \quad (13)$$

This effectively separates the high-energy gait signal from the low-energy noise tail without requiring prior knowledge of the sensor’s noise specifications.

4.3 Preprocessing Pipeline

Based on the diagnosis above, we applied the following preprocessing steps to clean and consolidate the data:

- **Adaptive Low-Pass Filtering:** We applied a zero-phase Butterworth low-pass filter of order 5. Unlike a standard fixed-cutoff approach, we utilized the subject-specific cutoffs $f_c^{(s,p)}$ derived from our spectral analysis. This ensures maximal noise suppression (denoising) while preserving the morphological sharpness of heel-strikes and toe-offs (see Figure 1).
- **Standardization:** To handle amplitude variations between sensors (e.g., acceleration vs. angular velocity), each channel was standardized (zero mean, unit variance) individually. This prevents channels with larger numerical ranges from dominating the ℓ_2 reconstruction error term in the PerCDL objective function.
- **Data Structure:** After preprocessing, the dataset was organized into a tensor $X \in \mathbb{R}^{S \times N \times P}$, where $S = 19$ represents the number of subjects, $P = 9$ the number of channels, and N the length of the continuous recording.

5 Results

In this section, we evaluate the performance of our PyTorch implementation of the PerCDL framework on the preprocessed clinical gait dataset. We structure our analysis by first examining the baseline performance of standard Convolutional Dictionary Learning (PopCDL), then demonstrating how the PerCDL framework enhances these results through personalization, and finally providing a quantitative comparison.

5.1 Experimental Setup

We trained both models on the full cohort of 19 subjects using the preprocessed tensor X . The hyperparameters were chosen to align with the biomechanical nature of the signal:

- **Dictionary Size:** $K = 2$ atoms. This choice was biologically motivated by the bipedal nature of human locomotion. Since the gait cycle consists of distinct movement patterns for the

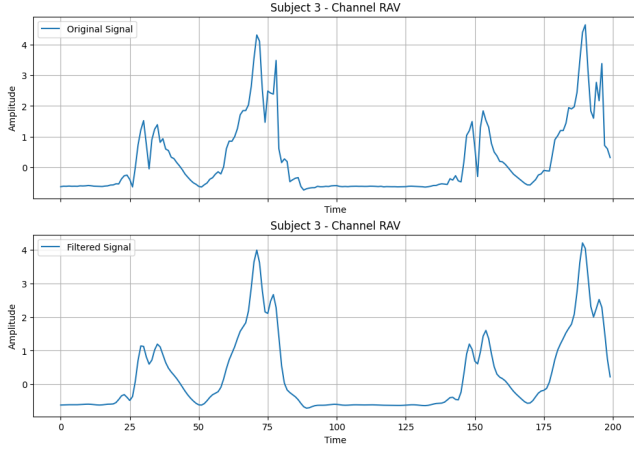


Figure 1: Signal before (top) and after (bottom) adaptive low-pass filtering, showing noise suppression with preserved morphology.

left and right legs (alternating stance and swing phases), we set $K = 2$ to allow the model to learn a specific primitive for the right foot displacement and another for the left foot displacement.

- **Atom Length:** $L = 50$ samples (0.5 seconds), corresponding to approximately half a gait cycle.

5.2 Standard PopCDL Analysis

We first applied standard Convolutional Dictionary Learning (PopCDL) to establish a baseline. This method assumes a rigid, shared dictionary across all subjects, treating individual variations as noise or residue.

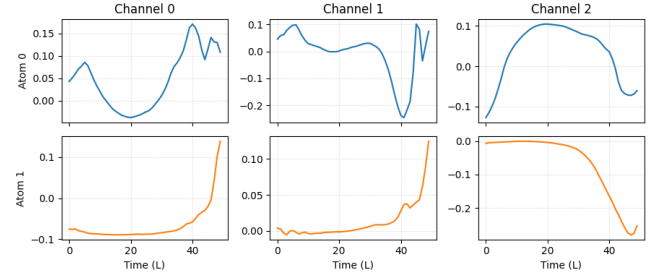
5.2.1 Global Atoms and Activations. Figure 2 (top) displays the global atoms learned by the model. Atom 0 and Atom 1 show distinct structural features, successfully capturing the complementary phases of the gait cycle. The sparse activations (Figure 2, middle) confirm this separation: the two atoms activate in an alternating, rhythmic pattern consistent with the periodic nature of walking.

5.2.2 Baseline Reconstruction. The reconstruction capabilities of PopCDL are shown in Figure 2 (bottom). While the model captures the fundamental periodicity of the signal, it struggles with temporal precision. The "rigid" atoms cannot stretch or compress to match the varying stride durations of the subject, leading to phase misalignment and an inability to fit sharper peaks.

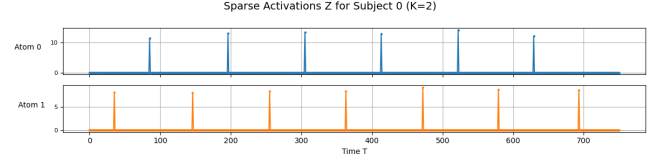
5.3 PerCDL and Personalization

We then introduced the Personalization layer (PerCDL), allowing the global atoms to be warped via a learned deformation field specific to each subject.

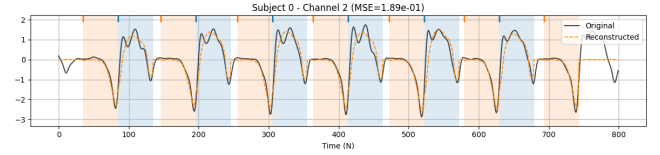
5.3.1 Warped Atoms. The effect of this personalization is visualized in Figure 3. The global template (dashed gray) acts as a "mean shape." The personalized atoms (solid lines) are derived from this template but exhibit distinct temporal shifts and compressions. For Subject 0, the model learns to slightly compress the atom duration



(a) Global atoms learned by standard CDL ($K = 2$).



(b) Sparse activations showing alternating firing of atoms.



(c) Baseline reconstruction (Subject 0, Channel 2). Note the phase mismatch.

Figure 2: Standard PopCDL Results. The model learns interpretable atoms (a) that activate rhythmically (b), but the rigid reconstruction (c) fails to capture specific temporal variations.

for Channel 2, validating the hypothesis that the algorithm adapts to the subject's specific cadence.

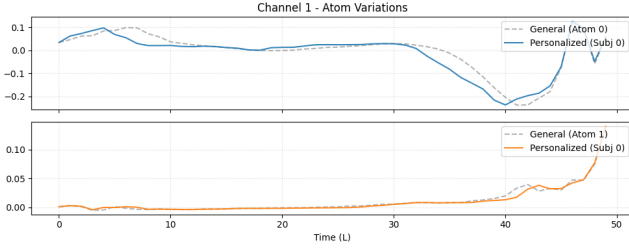
5.3.2 Improved Reconstruction. This adaptability directly translates to reconstruction quality. As shown in Figure 4, the PerCDL model (red line) achieves a significantly tighter fit to the original signal than the standard model. The warping allows the atoms to align perfectly with the complex double-peak structures, effectively correcting the phase errors observed in the baseline.

5.4 Quantitative Comparison

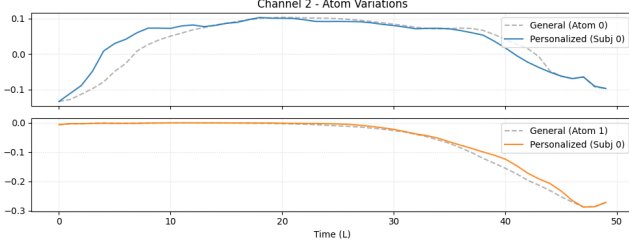
To robustly assess the performance of the proposed method, we conducted a comparative analysis between the standard CDL (PopCDL) and the Personalized CDL (PerCDL) frameworks across the entire cohort of 19 subjects. Both models were trained for 50 iterations ($n_{iters} = n_{perso} = 50$) to ensure convergence.

Figure 5 illustrates the average Mean Squared Error (MSE) per channel. On average, the PerCDL framework achieves a **7% reduction in reconstruction error** compared to the standard approach. However, the gain is non-uniform across channels:

- **High Variability Channels:** Channels such as Ch7 and Ch4 show significant improvements of **16.2%** and **16.0%** respectively. These likely correspond to axes of motion where individual anatomical differences (e.g., stride length, foot



(a) Atom 0 Variations (Channel 1)



(b) Atom 1 Variations (Channel 2)

Figure 3: Personalization Mechanism. The global template (gray dashed) is warped (solid color) to fit Subject 0’s specific timing.

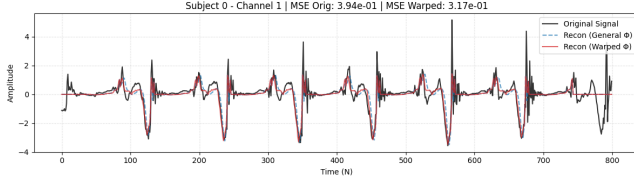


Figure 4: PerCDL Reconstruction (Subject 0, Channel 1). The warped atoms (red) capture signal details and timing nuances missed by the rigid global dictionary (blue dashed).

rotation) introduce substantial temporal variability that the rigid CDL model fails to capture.

- **Stable Channels:** Conversely, Channel 0 shows negligible improvement (**0.0%**). This suggests that the signal dynamics in this dimension are highly conserved across the population, making the "global" template sufficient for accurate reconstruction without the need for personalized warping.

This differential performance validates our hypothesis: PerCDL is most effective where biological variability is highest, while reverting to standard CDL performance where the signal is globally consistent.

5.5 Discussion & Clinical Perspectives

Beyond signal reconstruction, PerCDL provides a powerful framework for our “Immersion à l’Hôpital” project by treating the warping coefficients $A \in \mathbb{R}^{S \times K \times M}$ as interpretable digital biomarkers. Unlike raw data, these coefficients provide a compact representation of subject-specific deviations, such as temporal asymmetry or compensatory micro-deformations.

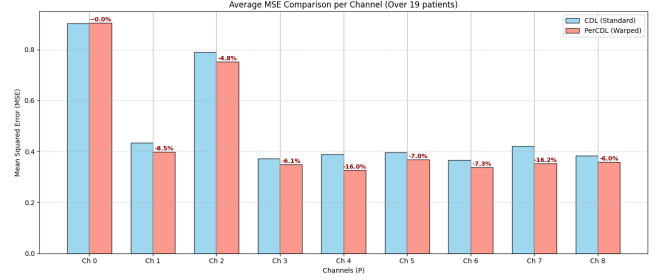


Figure 5: Average MSE Comparison per Channel (Cohort of $S = 19$). PerCDL (red) outperforms standard CDL (blue) by an average of 7%, with gains reaching up to 16.2% on highly variable channels (e.g., Ch7).

We propose to leverage this in two ways: first, by using these coefficients as input features for classifiers (e.g., SVM, PCA) to identify a latent “signature of pain” that distinguishes pathological gait. Second, we utilize the generative nature of PerCDL to synthetically manipulate these coefficients. This allows us to visualize exactly how specific pathological parameters alter the gait cycle, offering clinicians an explainable diagnostic tool rather than a black-box prediction.

References

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